Controlled Continual Learning as Constrained Representational Motion

Local research note

Generated from Colab plasticity-audit artifacts

Abstract

This note reports one focused toy continual-learning result. A small transformer is trained on an old world, then receives staged new and changed facts. Naive sequential training forgets protected old behavior. A projection-only Invariant-Tangent update also fails because local projection does not correct accumulated drift. The successful mechanism is **Invariant-Tangent CL + restore**: project the new-learning gradient into the tangent space of protected behavior and geometry, then add a small bounded restore field toward the protected manifold. In one Colab run, this mechanism preserves old and guard behavior, learns changed and new behavior, suppresses obsolete answers, and keeps residual geometry closer to the base model than naive or joint-from-scratch references. This is promising evidence, not a robustness proof.

1 What Is Being Claimed

The supported claim is deliberately narrow:

$$\text{controlled tangent update} + \text{bounded restore}$$

can work in this toy staged CL setting. It is not yet a claim of scale, multi-seed robustness, autonomous role discovery, or a full continual-learning solution.

The failed run and successful run use the same staged protocol. The key change is:

Run	Update
Projection-only failure	$g_{\text{update}} = g_{\text{tangent}}$, restore strength 0.0
Successful run	$g_{\text{update}} = g_{\text{tangent}} + 0.05 g_{\text{restore}}$

2 Mechanism

For new examples N_t, compute:

$$g_N = \nabla_\theta L_N(\theta_t).$$

Protected behavior and geometry define invariant functions $c_i(\theta)$: preserve behavior KL, guard behavior KL, residual-state geometry, role centroid geometry, feature centroid geometry, and pairwise separation geometry. Their Jacobians form:

$$A_t = \begin{bmatrix} \nabla_\theta c_1(\theta_t) \\ \nabla_\theta c_2(\theta_t) \\ \vdots \\ \nabla_\theta c_m(\theta_t) \end{bmatrix}.$$

The tangent gradient removes the locally dangerous component:

$$g_{\text{tangent}} = g_N - A_t^\top (A_t A_t^\top + \rho I)^- 1 A_t g_N.$$

Projection alone was not stable. The successful update adds a bounded restore field:

$$g_{\text{update}} = g_{\text{tangent}} + \alpha_{\text{restore}} g_{\text{restore}},$$

where:

$$g_{\text{restore}} = \nabla_\theta \left[L_{\text{preserve}} + L_{\text{guard}} + L_{\text{geometry}} \right].$$

Interpretation: the tangent term preserves plasticity by learning through directions compatible with protected invariants. The restore term prevents finite-step drift from accumulating away from the protected manifold.

3 Behavior Results

Method	Preserve	Guard	Changed	New	Compose	Suppress obsolete
Base model	1.000	1.000	0.000	0.000	0.333	0.000
Naive sequential	0.000	0.000	0.000	0.300	0.167	1.000
Invariant-Tangent CL, no restore	0.000	0.000	0.000	0.300	0.167	1.000
Invariant-Tangent CL + restore	1.000	1.000	1.000	1.000	1.000	1.000
Joint-from-scratch reference	1.000	1.000	1.000	1.000	1.000	1.000

Table 1: Success scores. Higher is better for every column; suppress obsolete is inverted so 1 means the obsolete old answer is no longer produced.

Method	Preserve	Guard	Changed	New	Compose
Naive sequential	5.9348	5.5818	5.9220	2.6712	3.7792
Invariant-Tangent CL, no restore	8.1709	8.4551	7.3757	4.3208	6.1792
Invariant-Tangent CL + restore	0.0001	0.0001	0.0026	0.0020	0.0014

Table 2: Raw behavior losses. Projection-only learns the new stage but loses protected behavior. The restore run keeps all target categories near zero loss.

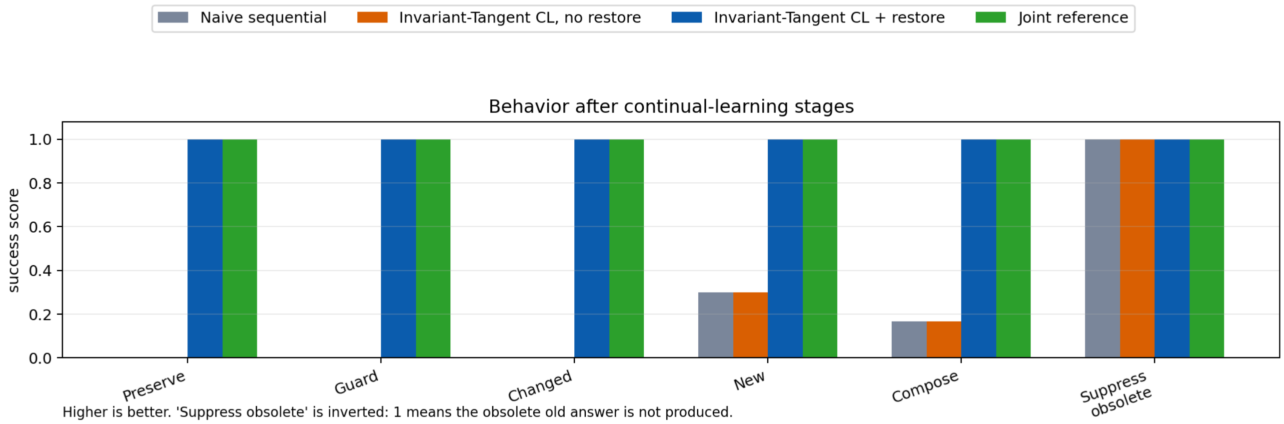

Figure 1: Clean behavior view rebuilt from JSON with corrected method labels.

4 Residual Geometry

Comparison	Drift relative ↓	CKA ↑	Rank delta
Naive vs base	0.575	0.672	-0.003
Invariant-Tangent CL, no restore vs base	0.569	0.628	-1.512
Invariant-Tangent CL + restore vs base	0.393	0.861	8.310
Joint reference vs base	0.820	0.550	8.513

Table 3: Mean residual geometry relative to the base model. **Invariant-Tangent CL + restore** has lower residual drift and higher CKA than naive sequential training, while joint training learns everything by building a more different geometry.

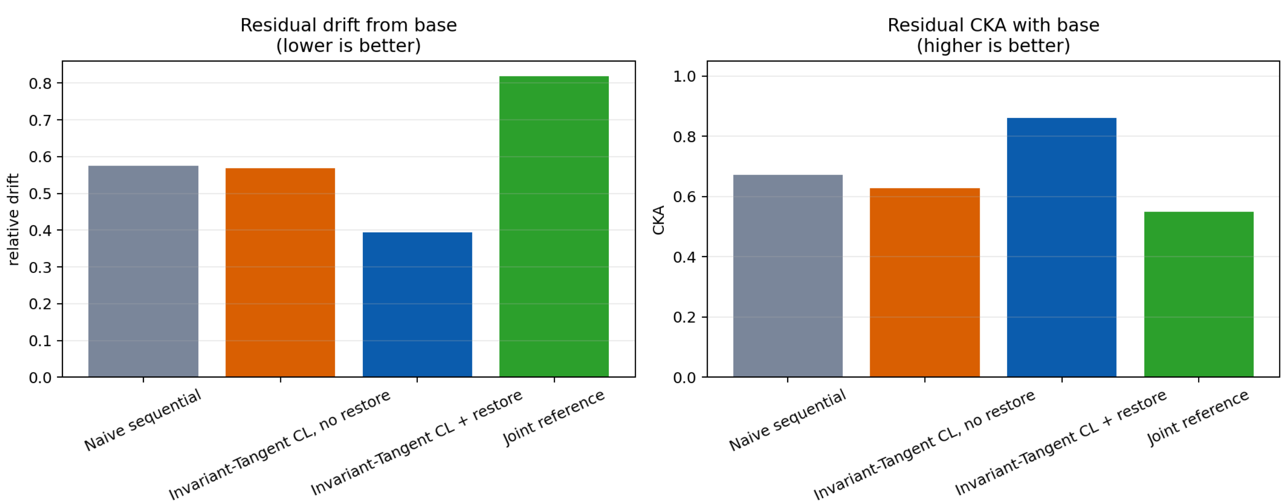

Figure 2: Residual geometry comparison rebuilt from JSON.

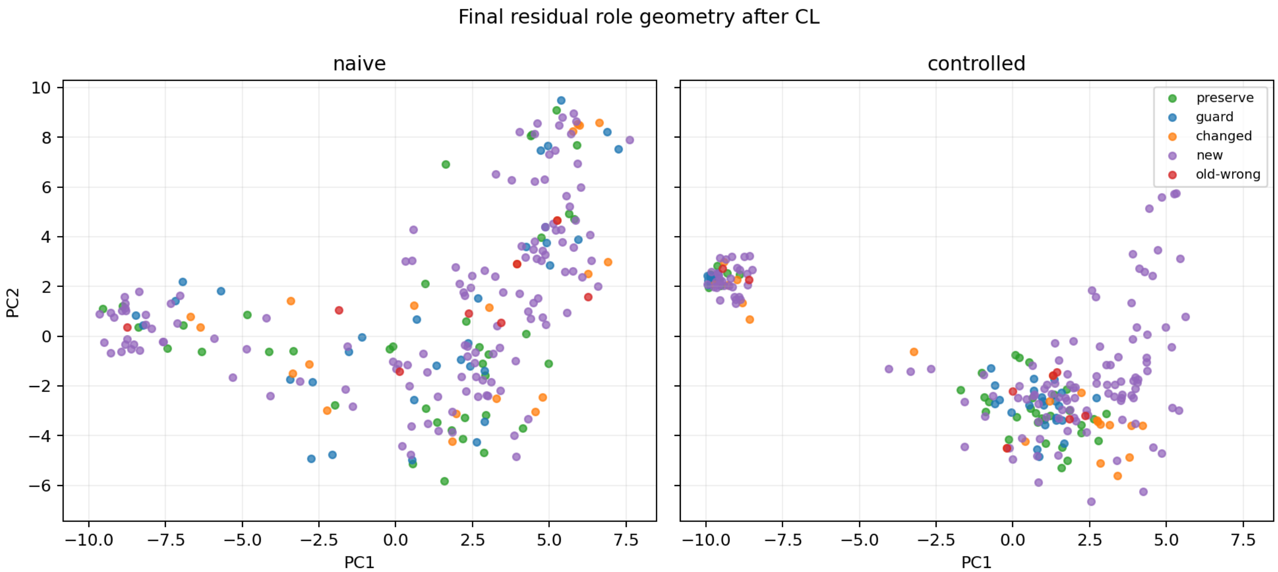

Figure 3: Raw PCA figure from the successful run. The right panel title is the experiment's internal key "controlled"; in this JSON that key means **Invariant-Tangent CL + restore** because the run used projected-invariant-tangent mode with restore strength 0.05. The bitmap is included as generated, not relabeled.

5 Role And Feature Geometry

Comparison	Centroid drift ↓	Centroid cosine ↑	Group CKA ↑	Separation drift ↓
Naive vs base	4.272	0.532	0.684	0.142
Invariant-Tangent CL + restore vs base	2.644	0.794	0.766	0.159
Joint reference vs base	6.285	-0.008	0.711	0.259

Table 4: Role geometry. The restore run improves centroid drift and centroid cosine versus naive, while separation drift is mixed.

Comparison	Centroid drift ↓	Centroid cosine ↑	Group CKA ↑	Separation drift ↓
Naive vs base	4.399	0.543	0.647	0.147
Invariant-Tangent CL + restore vs base	2.615	0.819	0.697	0.224
Joint reference vs base	6.257	0.005	0.655	0.216

Table 5: Feature geometry. The restore run preserves feature centroids much better than naive; separation metrics are not uniformly better.

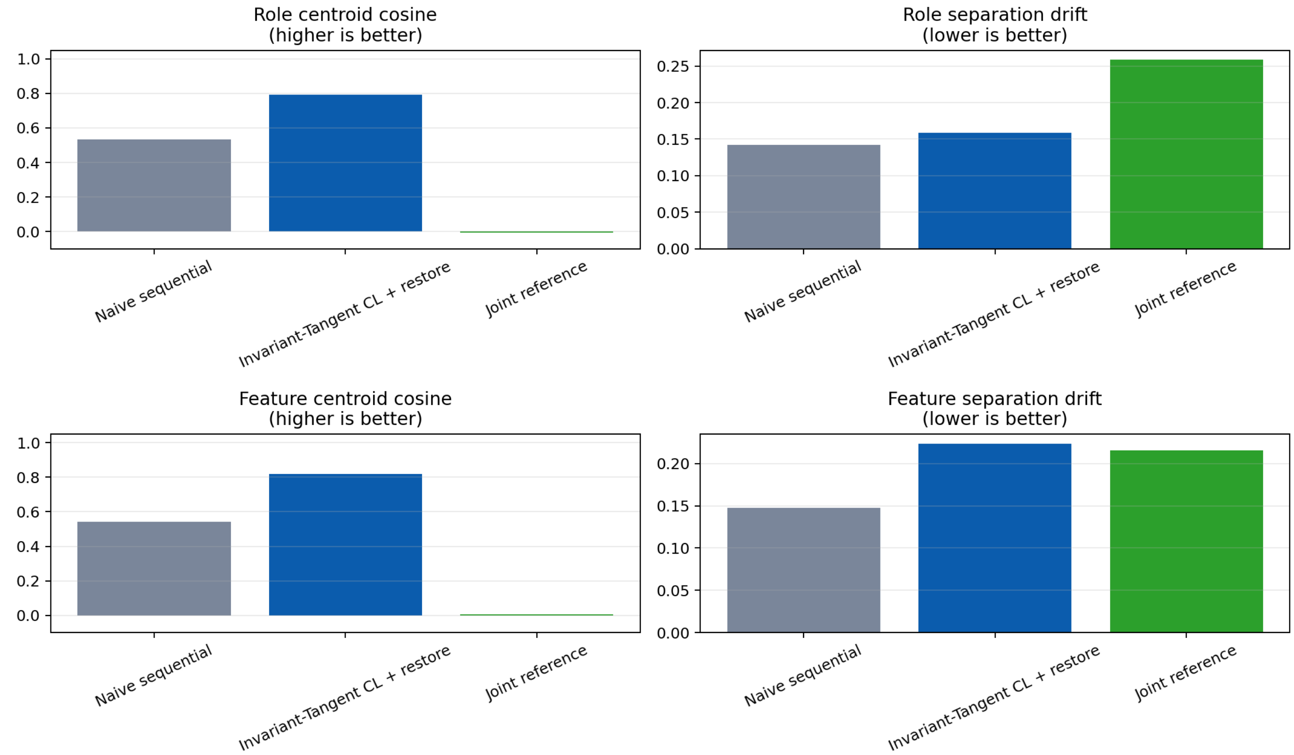

Figure 4: Role and feature geometry metrics. This avoids the overclaim that the method wins every geometry metric; the strongest result is reduced global residual drift and better centroid preservation versus naive.

6 Plasticity And Restore

Run/stage	Safe/raw	Final/raw	Removed	Effective rank	Redundancy	Memory
Invariant-Tangent CL, no restore, stage 1	0.9987	0.9987	0.0480	3.026	0.496	9
Invariant-Tangent CL, no restore, stage 2	0.9982	0.9982	0.0588	1.106	0.941	18
Invariant-Tangent CL, no restore, stage 3	0.9930	0.9930	0.1169	1.081	0.948	27
Invariant-Tangent CL, no restore, stage 4	0.9981	0.9981	0.0614	1.072	0.940	36
Invariant-Tangent CL + restore, stage 1	0.9995	1.4272	0.0273	5.282	0.120	9
Invariant-Tangent CL + restore, stage 2	0.9988	1.4413	0.0466	1.107	0.941	18
Invariant-Tangent CL + restore, stage 3	0.9972	1.0591	0.0739	1.046	0.950	27
Invariant-Tangent CL + restore, stage 4	0.9990	1.2120	0.0436	1.078	0.940	36

Table 6: Plasticity audit. Safe/raw stays near one, so plasticity does not collapse. The restore run has final/raw above one because the corrective field adds motion back toward protected behavior and geometry.

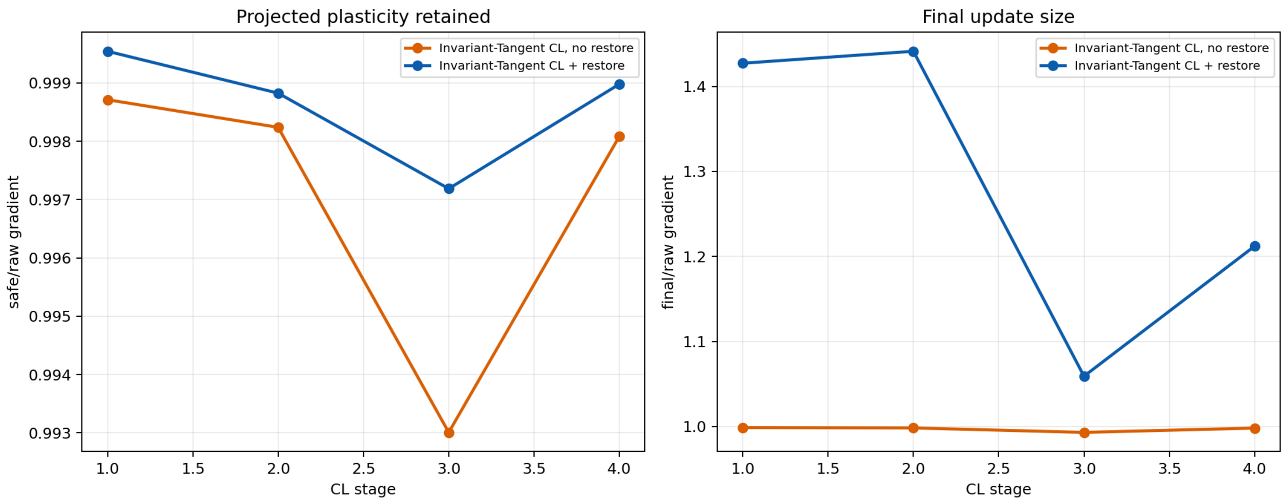

Figure 5: Clean plasticity comparison. Projection-only leaves plasticity but does not preserve behavior; restore adds the stabilizing correction.

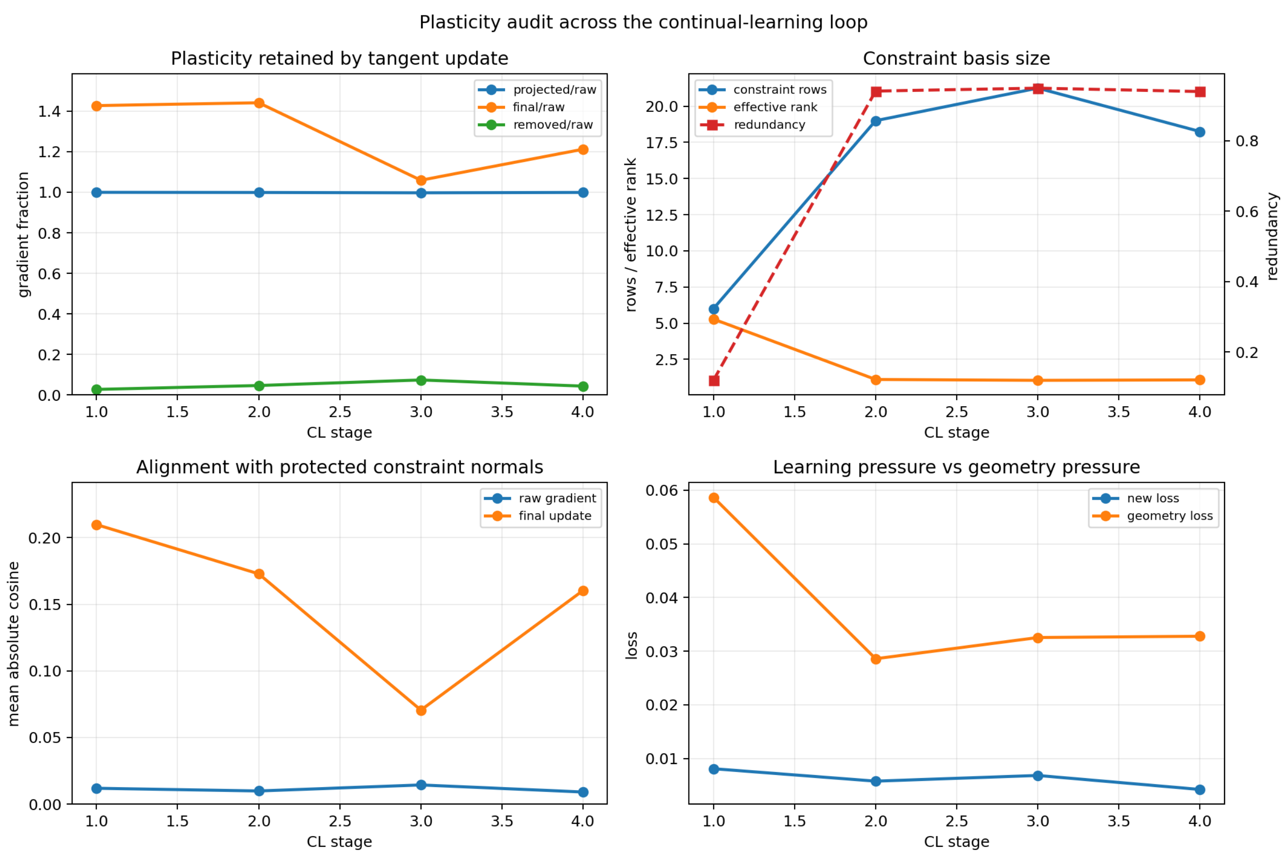

Figure 6: Raw plasticity audit from the successful run. It shows safe/raw near one across all four stages and final/raw above one due to restore.

7 What This Does Not Prove

This result is one strong run. It does not prove:

- restore strength 0.05 is the unique or optimal value;
- the mechanism is robust across seeds;
- the role assignment can be fully discovered without external labels;
- the method scales to large models or long open-ended streams;
- anchor memory remains bounded over much longer loops.

The next robustness test is a restore-strength sweep and a small seed sweep. The current result is enough to justify the mechanism as a promising direction, not enough to present it as solved continual learning.

8 Short Conclusion

The useful lesson is not simply "add distillation." Projection-only failed, which shows that local tangent constraints are insufficient by themselves. The working mechanism is the combination:

$$\boxed{g_{\text{update}} = g_N - A_t^\top (A_t A_t^\top + \rho I)^- 1 A_t g_N + \alpha_{\text{restore}} g_{\text{restore}}}$$

This keeps the new-learning direction mostly plastic while using restore to stay near protected behavior and geometry. In the reported toy run, that was enough to avoid catastrophic forgetting and learn the staged updates.